Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

BACKUP POWER ADVICE

Harmony @Huailan Ecolodge: keeping the lodge, the water pump and hot water running during a power cut

Existing system: 10 kWp solar panels, 10 kW three-phase hybrid inverter, about 15 kWh lithium battery

The short version

- **The lodge and the water pump can be backed up** with the system you already have.
- **The hot water heater, as it is today, cannot.** It is too big for the inverter (about 11–12 kW when it switches on). During a power cut it would shut the whole system down.
- **Our recommendation:** replace it with a heat-pump water heater and a hot-water tank (Option A). Then the lodge, the pump and hot water all run during a power cut, and your monthly bill goes down too.

You asked us to keep the whole lodge, including the hot water heater and the water pump, working when the grid power goes off. Here is what is possible.

1. What your system can supply during a power cut

When the grid is off, everything runs from your inverter and battery. The inverter has three limits:

- **10 kW in total**, across all three electrical phases together.
- **5 kW on any single phase.**
- A short burst of about twice that power, for **up to 10 seconds**. This helps motors such as pumps to start.

Your battery stores about 15 kWh. That is enough to run the normal lodge (about 0.5–1 kW) through a typical evening and night.

2. What we measured at your lodge

Load	What it needs	Fits on backup today?
Lodge lights, fridges, Wi-Fi, sockets	About 0.5 kW at night, 1 kW on average, 3–6 kW at busy times	**Yes**
Water pump	To be checked on site (usually 1–2.2 kW, with a short starting surge)	**Very likely yes**
Hot water heater	About **11–12 kW**: around 6 kW on each of two phases, for a few minutes, about once a day (early morning and late evening)	**No**: above both the 10 kW total and the 5 kW per-phase limit

Moving the heater to another phase or another outlet does not solve this. The limit is the size of the inverter, not the wiring.

3. Your options for backed-up hot water

Option	What we do	Good points	Things to know
A. Heat-pump water heater + storage tank (recommended)	Replace the large heater with a heat-pump water heater (about 1–1.5 kW) and an insulated hot-water tank. The tank is heated slowly, mostly by daytime solar.	Works on backup with your existing inverter and battery. Uses about one-third of the electricity of a normal heater, so it lowers your bill. It also uses the free solar that is wasted today.	Needs space for the tank and heat pump. The tank must be sized for your busiest shower times.
B. Smaller heating elements + storage tank	Replace the large heater with smaller elements (no more than about 3–3.5 kW each), spread across the three phases, heating a storage tank.	Works on backup. Costs less than a heat pump.	Uses about 3 times more electricity than a heat pump, so the battery empties faster during a power cut.
C. Keep the current heater, make the system bigger	Add a second 10 kW inverter working together with the first, plus more battery storage.	No change to your hot water system or habits.	By far the most expensive. The battery would also need to grow, because your current battery can only supply about 10 kW. Every shower during a power cut uses a lot of stored energy.

4. The water pump

- We will check its size and starting current on site. Most pumps up to about 2.2 kW start without problems on backup.
- For a larger pump or a three-phase pump, we add a soft starter or a variable-speed drive. This makes it start gently and avoids overloading the inverter.
- A raised water tank keeps water flowing even while the pump is off.

5. Our recommended plan

1. **Site visit.** We check the heater type and size, the pump, your main breaker, and which circuits are on which phase.
2. **Move all lodge circuits and the pump to the backup side** of the inverter, balance the three phases, and add a manual bypass switch so the lodge can still be fed from the grid during servicing.
3. **Hot water:** Option A (recommended) or Option B. Option C only if you prefer to keep the current heater.
4. **Battery settings:** we keep a reserve of about 40% at night, so there is energy ready for a power cut. During the day the battery can be used down to 25% to save on your bill.

What you will get with Option A

In a normal evening or night-time power cut, lights, fridges, Wi-Fi, sockets, the water pump and hot water all keep working from the solar battery. For very long cuts (a day or more), a small generator can be connected as an extra safety net.

We will quote Options A, B and C after the site visit. Call 063 151 6694 or 089 632 5829, LINE: solaress1. **Prepared by Solar ESS Working Group**